



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number :[BITS F431]
Course Title :[Flexible Manufacturing Systems]
Instructor-in-charge :[Tufan Chandra Bera]
[Team of Instructors] :[Tufan Chandra Bera, tcbera@pilani.bits-pilani.ac.in, 2226]

1. Course description:

Introduction of CAD/CAM systems, overview of FMS, system hardware and general functions, material handling system, work holding systems, cutting tools and tool management, physical planning of system, software structure functions and description, cleaning and automated inspection, communications and computer networks for manufacturing, quantification of flexibility, human factors in manufacturing, FMS and CIM in action (case studies), justification of FMS, modelling for design, planning and operation of FMS.

2. Scope and Objective:

In recent years the manufacturing industries are facing problems of survival. They find it difficult to maintain a long term competitive advantage due to uncertainty in environmental conditions. The ideal production system for such environment is Flexible Manufacturing Systems (FMS), which are considered to meet the demand for a variety of products with improvement of productivity in mid-volume manufacturing. This course will highlight the fundamentals of FMS and modelling for design, planning and control of FMS.

3. Course learning outcomes:

CLO1. Comprehend importance of adaptability of changes in product type and quantity in industry.

CLO2. Understand how labor cost can be reduced by automated the manufacturing system using FMS.

CLO3. Comprehend CNC technology and role of CAM in FMS.

4. Text books and references:

TB. Mikell P Groover, “Automation, Production Systems and CIM”, 4th Edition, Pearson Education, 2016.

RB1. Rao P.N., “CAD/CAM, Principles and Applications”, 3rd Edition, TMH.

RB2. Talavage J. & Hannam, R.G. “Flexible Manufacturing Systems in Practice”, Marcel Dekker Inc, 1988.

RB3. Greenwood N.R, “Implementing Flexible Manufacturing Systems”, Macmillan Education, 1st Edition, 1988.

RB4. William W.L, “FMS Cells and Systems”, PHI, 1991.

RB5. Ranky P.G, “Design and Operation of FMS”, IFS Publications, 1983.

5. Course Format & Weekly Workflow

weekly schedule for 14 weeks with [3] lectures per week

Wk no	Module no	Description	Ref	Learning Outcome
1-2	1	Introduction to Production Systems.	TB - Ch.1	Familiarization of manufacturing System
	2	Manufacturing Operations,	TB - Ch 2,	Overview of Manufacturing
		Manufacturing Models and Metrics.	TB - Ch.3	
3-6	3	Introduction to Automation	TB - Ch. 4	Familiarization of Automation and Control Technologies
		Industrial Control Systems	TB - Ch. 5	
		Hardware components for Automation and Process Control	TB - Ch. 6	
		Numerical Control	TB - Ch. 7	
		Industrial Robotics	TB - Ch. 8	
		Discrete Control using PLC and PC	TB - Ch. 9	
7-9	4	Material Transport Systems	TB - Ch.10	Familiarization of Material Handling and Identification Technologies
		Storage Systems	TB - Ch. 11	
		Automatic Identification and Data Capture	TB - Ch. 12	
10-13	5	Introduction to Manufacturing Systems	TB - Ch. 13	Familiarization of Manufacturing Systems and FMS
		Single Station Manufacturing Cells	TB - Ch. 14	
		Manual Assembly Lines	TB - Ch. 15	
		Automated Production Lines	TB - Ch. 16	
		Automated Assembly Systems	TB - Ch. 17	
		Cellular Manufacturing	TB - Ch. 18	
		Flexible Manufacturing Systems	TB - Ch. 19	
14	6	Advanced topics in FMS	Digital contents available in internet	Familiarization of advancement of FMS towards industry 4.0
		Advanced topics in FMS	Internet	
		Advanced topics in FMS	Internet	

6. Evaluation Scheme:

Component	Duration (Min.)	Weightage (%)	Date & Time	Nature of component (Closed Book/ Open Book)
Midsem Test	90 Min.	30	07/10/2026 FN1	Open Book
Assignments / Case Studies	Semester long	30	To be announced later	Open Book Presentation & Viva voce
Comprehensive Examination	180 Min.	40	10/12/2026 FN	Closed Book

Criterion for NC: The minimum passing marks of this course will be determined based on the 40% of the average score of the class strength. This criterion will be considered as passing marks for valid grade in this course.

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

8. Chamber consultation hours: To be announced in the class

9. Makeup Policy: [Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.]

Tufan Chandra Bera
Instructor-in-charge
[BITS F431]